Deprescribing

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INTRODUCTION

Deprescribing is an essential part of good prescribing and is inherently linked to related activities, such as medication reconciliation, to ensure safe and effective use of medications. This process requires attention, time, and in many cases special skills and knowledge. This includes technical knowledge, such as optimal down-titration schedules, as well as competencies in shared decision making, communication, and managing health systems in a medical culture that historically has been more oriented toward adding medications than stopping them. A more general discussion of drug prescribing in older adults is provided separately. (See "Drug prescribing for older adults".)

Deprescribing is most commonly employed in older adults, people with multiple chronic conditions, and people near the end of life. However, deprescribing can be appropriate in any patient taking medication. (See 'Which patients and which medications should be targets for deprescribing?' below.)

This topic will provide information on approaches to deprescribing, including common barriers and how to address them. Related issues regarding discontinuing antihypertensive and antidepressant agents are discussed in detail elsewhere. (See "Tapering and discontinuing antihypertensive medications" and "Antidepressant discontinuation syndrome and discontinuing antidepressants in adults".)

DEFINITION

The term "deprescribing" refers to a process of medication withdrawal, supervised by a health care professional, with the goal of managing polypharmacy and improving outcomes [1]. This can encompass efforts to comprehensively review a patient's medication list and systematically discontinue or reduce the dose of all medications with an unfavorable balance of benefits and harms, as well as efforts focused on specific types of high-risk medication. What distinguishes deprescribing from traditional approaches to pharmaceutical care is that it encourages a systematic and proactive approach. Deprescribing focuses on being proactive to address medication-related problems that have not previously been identified or satisfactorily managed and to prevent future problems.

GOALS OF DEPRESCRIBING

The overall goal of deprescribing is to improve patient important outcomes. Patients can vary in their experience of potential benefits and harms from medication use, how they value these outcomes, and their attitudes toward medication withdrawal. Therefore, the goals of deprescribing can vary between patients. For example, for a patient who has recently suffered an injurious fall, deprescribing efforts may focus on reducing use of medications that confer increased fall risk. In another patient who is troubled by taking multiple medications dosed multiple times per day, efforts may focus on reducing medication regimen complexity and burden.

Common goals for deprescribing include reducing overall medication burden, reducing the risk of specific geriatric syndromes such as falls and cognitive impairment, and improving global health outcomes such as hospitalization and death. Ultimately, all these goals relate to improving quality of life. Specific goals can include:

- Reducing medication burden – Overall, deprescribing interventions are generally effective in reducing the number of medications and improving appropriateness of medication use [2]. Careful medication review and aggressive discontinuation can lead to reduction in up to 39 percent of medications used [3], including reducing use of potentially inappropriate medications by up to 30 to 60 percent or more [4-6]. Reducing medication burden may improve adherence to remaining medications [7].

- Reducing risk of falls – Many medications increase fall risk among older adults, including benzodiazepines and benzodiazepine receptor agonists (eg, zolpidem and zopiclone), antidepressants, antipsychotics, and strongly anticholinergic medications [8-11]. In nursing home settings, deprescribing programs using broad-based medication review have been shown to reduce fall risk by 24 percent [4]. Efforts to reduce use of psychotropic medications have also reduced the rate of falls, albeit with highly variable results across studies [12,13]. However, while deprescribing of fall-risk increasing drugs and other potentially inappropriate medications appears clinically sensible, several studies have found that interventions to deprescribe these medications have not meaningfully affected the rate of falls [2,6,14-16]. (See "Falls: Prevention in community-dwelling older persons", section on 'Medication modification' and "Falls in older persons: Risk factors and patient evaluation",

section on 'Medication use' and "Falls: Prevention in nursing care facilities and the hospital setting", section on 'Medication use'.)

- Improving and/or preserving cognitive function – Anticholinergic medications, sedative-hypnotics including benzodiazepines and benzodiazepine receptor agonists, and use of multiple psychotropic medications can negatively affect cognition [9]. However, several studies have found no meaningful improvement in cognitive function in people with and without dementia following a deprescribing intervention [17,18]. By contrast, a small number of studies have found a benefit when deprescribing psychotropic medications [19-21].

- Reduce risk of hospitalization and death – Overall, there are mixed findings on the impact of deprescribing interventions on hospitalization and death [2,22]. In vulnerable older adults residing in nursing homes, trials of interventions incorporating whole-regimen review and deprescribing reduced hospitalization by 36 percent and death by 26 percent [4]. In ambulatory settings, a systematic review found that interventions using whole-medication review had no impact on hospitalization but did reduce mortality by 26 percent [6], although methodologic challenges may reduce confidence in this finding [2,22,23].

In addition to these specific goals of deprescribing, medication deintensification can be framed as part of good clinical practice, since all medications are potentially harmful, cost money, and add complexity and the potential for burden. Careful medication review often identifies multiple drug-related problems such as adverse reactions, lack of effectiveness, and burdensome treatment costs, some of which can be successfully addressed through medication withdrawal [24].

Many trials of deprescribing interventions have not identified a difference in clinical outcomes [25-28]. Reasons for this may include the failure of the intervention to robustly promote deprescribing (such that few medications are actually stopped) , a focus on stopping medications which were low-risk or not causing harm, intervening on lower-risk patient populations, and insufficient sample sizes and short-term follow-ups [29,30]. Such findings highlight the importance of focusing deprescribing efforts on people and medications where the potential yield is greatest. Moreover, deprescribing may improve less commonly-measured outcomes such as patient's medication burden, comfort with medications, quality of life, and costs. Finally, adverse drug withdrawal events (ADWEs) are uncommon with deprescribing and in many cases deprescribing is safe [31,32].

WHICH PATIENTS AND WHICH MEDICATIONS SHOULD BE TARGETS FOR DEPRESCRIBING?

Medication-related problems are extremely common in the general population of adults, and especially in older adults, who take a disproportionately high share of medications [24,33-36]. These problems include adverse drug effects, ineffectiveness, use of medications with no indication, excessive or inadequate dosing, use of potentially inappropriate medications, and nonadherence. Thus, careful medication review and consideration of deprescribing is appropriate for most people who chronically use medications.

Although opportunities for deprescribing should be considered for all individuals who use medications, certain risk factors can identify which patients are at highest risk for medication misadventures and thus may be highest priority for careful medication review and deprescribing where appropriate. Patient characteristics and medications which are good targets for deprescribing are presented in the table (table 1).

Patient characteristics — High-risk patient characteristics include:

- Polypharmacy – Use of multiple medications is the strongest risk factor for medication-related problems [37-43]. In a study of older veterans, those in the highest quartile of medication use had 6 to 12 times the risk of medication-related problems compared with those in the lowest quartile of medication use [38]. There is no specific threshold number of medications above which risk increases; each incremental medication confers additional risk [38]. However, evidence suggests that use of five or more medications is a useful benchmark for identifying older adults at higher risk [44].

- Multimorbidity – As the number of chronic conditions increases, the risk of drug-disease interactions (medications for one condition adversely affect a concurrent condition) also increases substantially [38,39,41].

- Renal impairment – Decreased kidney function increases the risk for adverse drug events, especially if there is inappropriate medication dosing [38,39,43,45].

- Multiple prescribers and transitions of care – Miscommunication about medications can commonly occur when a patient has multiple prescribers or transitions from one care setting to another (eg, from hospital to a residential care facility or home). This can lead to inadvertent continuation of medications that should have been stopped either because they are causing documented harms or lack ongoing indications for use [46,47].

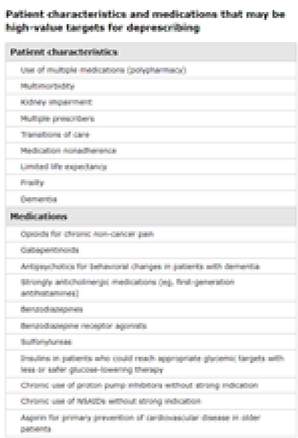
- Medication nonadherence – Nonadherence has many causes, including dissatisfaction with medications, difficulty with medication use, and real or perceived adverse effects and/or lack of benefit. Optimizing care depends on the underlying problems and clinical scenario, but for some patients deprescribing may help them achieve the goals they value.

- Limited life expectancy – Many patients with limited life expectancy prefer to reduce their medication burden. With a shift in goals of care from prolonging life to improving the quality of life, the goals of deprescribing may also shift, with different medications becoming key targets for deprescribing [48,49]. Preventive medications are often continued up until the end of life, despite changed care goals and the patient being unlikely to benefit in their remaining lifespan [50-53]. Examples of such therapies include statins, agents for aggressive control of hyperglycemia, and bisphosphonates [54-56]. In a landmark trial, discontinuation of statins in people with less than 12 months to live did not alter time to death compared with continuation, and those in the discontinuation group had better quality of life [55]. On the other hand, certain high-risk medications, such as opioids, sedatives, and anticholinergics, may be reasonable to continue because of their symptomatic benefits near the end of life and lower concerns about long-term risks.

- Older age – Advanced age is not itself an independent risk factor for medication-related problems [57]. However, older adults are often at high risk of such problems due to the increased incidence of polypharmacy, multimorbidity, transitions of care, changes in pharmacokinetics and pharmacodynamics, and other risk factors [58].

- Frailty and dementia – People with frailty and/or dementia may have different and changing goals of care, and there may be different opportunities to use medications safely and effectively [59,60]. Moreover, older adults with dementia receive potentially inappropriate medications at higher rates than the general population [61,62] and have high rates of central nervous system (CNS)-active polypharmacy [63]. Frailty and dementia may increase the risk of medication-related harms and change goals of care. In some cases, adverse drug effects may precipitate or worsen these syndromes (for example, due to medication-induced fatigue or cognitive changes) [9,64]. Caregivers may be engaged to be a facilitator in the deprescribing process; however, they also need support, as surrogate decision makers may feel pressure and guilt about making deprescribing decisions [59].

Medications — Characteristics of the medication regimen which should prompt consideration of deprescribing are presented in the table (table 1) and include:


Patient characteristics and medications that may be high-value targets for deprescribing

Patient characteristics and medications that may be high-value targets for deprescribing
Patient characteristics
Use of multiple medications (polypharmacy)
Multimorbidity
Kidney impairment
Multiple prescribers
Transitions of care
Medication nonadherence
Limited life expectancy
Frailty
Dementia
Medications
Opioids for chronic non-cancer pain
Gabapentinoids
Antipsychotics for behavioral changes in patients with dementia
Strongly anticholinergic medications (eg, first-generation antihistamines)
Benzodiazepines
Benzodiazepine receptor agonists
Sulfonylureas
Insulins in patients who could reach appropriate glycemic targets with less or safer glucose-lowering therapy
Chronic use of proton pump inhibitors without strong indication
Chronic use of NSAIDs without strong indication
Aspirin for primary prevention of cardiovascular disease in older patients

Patient characteristics and medications that may be high-value targets for deprescribing

- Use of potentially inappropriate medications for older adults – This term typically refers to consensus lists of medications such as the American Geriatrics Society Beers Criteria and the Screening Tool for Older Person's Prescriptions (STOPP) criteria [65,66]. These criteria identify medications that are often problematic for older adults, with a disadvantageous balance of benefits and harms for many older adults compared with pharmacologic and nonpharmacologic

Table 1 - larger image below

alternatives. Commonly used medications on these lists include sedative-hypnotics such as benzodiazepines and benzodiazepine receptor agonists, strongly anticholinergic medications, sulfonylureas such as glyburide, and chronic use of proton pump inhibitors (PPIs) and nonsteroidal antiinflammatory drugs (NSAIDs) in the absence of compelling indications (see "Drug prescribing for older adults", section on 'Potentially inappropriate medications'). The American Geriatrics Society has published consensus recommendations for nonpharmacologic and pharmacologic treatment strategies for older adults that clinicians can use in place of these potentially inappropriate medications (eg, strategies for managing insomnia without benzodiazepines or benzodiazepine receptor agonists, uncomplicated gastroesophageal reflux disease without long-term use of PPIs) [67].

- Other high-risk medications – Examples include (but are not limited to) insulins and aspirin (table 1). Although insulin treatment is often necessary, many adults with advanced age, multimorbidity, or functional decline receive overaggressive glucose-lowering treatment that yields few benefits and substantially greater harms compared with more permissive approaches [68]. Insulins are a leading cause of severe adverse drug events, so unnecessary use of these medications is particularly problematic [69]. Aspirin can be highly beneficial for people with known coronary or cerebrovascular disease, yet the risk of serious bleeding increases substantially with older age. (See "Aspirin in the primary prevention of cardiovascular disease and cancer", section on 'Patients ages 70 years and older'.)

APPROACH TO DEPRESCRIBING

Deprescribing is best thought of as a multistage process, rather than simply the concrete action of stopping a medication. Multiple steps are necessary to ensure that the process is patient-centered and achieves the best possible outcomes. For example, inadequate documentation and/or communication at the end of the process can result in inappropriate medications being restarted.

Stepwise approach — We approach deprescribing in three phases, adapted from published frameworks [70-72].

- **Phase 1** – The first step is to engage the patient and gather relevant information. Before considering which medications to start, stop, or change, it is critical to know what a patient is actually taking, if they are having problems with any of their medications, and how medication use fits into the larger picture of their health status, goals, and preferences. Different members of the health care team, especially pharmacists, can be invaluable resources for conducting these medication reviews [32].

- Compile a list of all regular and when required (prn) medications, including over-the-counter and complementary medications. Include the following information:
  - Dose and frequency of each medication
  - Duration of use
  - Indication
  - Patient's experience of each medication (ie, are they effective, difficult to take, or do they cause any adverse effects?)
  - Adherence
- Review patient goals of care, preferences, and values.
- Consider the patient's overall susceptibility to drug-induced harms, for example, frailty, cognitive impairment, or other geriatric syndromes such as falls.
- Engage the patient (and caregivers/family where appropriate) in a discussion of the deprescribing process. Connect with other health care professionals who may need to be consulted or could assist with the process.
- The Geriatric 5Ms can be a useful framework to guide person-centered care through considering Medications, Mind, Mobility, Multicomplexity and What Matters Most. Using the Geriatric 5Ms to guide conversations with patients and family can help with gathering necessary information and informing which drugs to deprescribe [73].

- **Phase 2** – The next step is to identify and decide on drugs to deprescribe. Each medication should be evaluated for its potential to be reduced in dose or discontinued, considering the balance of current and potential future benefits and harms. Look for medications that:
  - Have no valid or current indication (eg, the condition has resolved or the indication doesn't require long-term treatment, such as proton pump inhibitors [PPIs] for uncomplicated gastroesophageal reflux disease [GERD])
  - Are causing or contributing to a known or suspected adverse drug reaction, including geriatric syndromes which may be unrecognized adverse drug reactions (eg, reduced mobility due to muscle pain from statins, incontinence exacerbated by a diuretic)
  - Were started as a result of a prescribing cascade, a situation in which one medication is started to treat a side effect of another medication (eg, nonsteroidal antiinflammatory drug [NSAID] use leads to increased blood pressure which leads to prescription of a new antihypertensive medication) [74-77] (see "Drug prescribing for older adults")
  - Are generally high risk in the population (eg, anticholinergics in older adults) or contribute to a risk that would be more worrisome in a particular population (eg, drugs that may increase the risk of falls in someone with osteoporosis)

- Are ineffective
- Are used for a preventive indication in a patient with a life-limiting illness (eg, bisphosphonates in people near the end of life)
- Cause unacceptable treatment burden (eg, insulin in a person with dementia who is fearful of needles)
- Are used to treat the same (or similar) conditions as other medications. Medications can often be added for unresolved medical conditions, without considering the need for continuation of previous medications. This can lead to complex medication regimens and high risk of harm (eg, multiple pain medications)

Explicit criteria of potentially inappropriate medications such as the American Geriatrics Society Beers Criteria can be helpful, but all medications should be reviewed with assessment of the balance of benefits and risks for each individual person.

Use shared decision making in deciding which medications to stop or reduce to ensure that patients' opinions and goals are recognized. (See 'Shared decision making' below.)

- **Phase 3** – The final steps involve planning, implementation, monitoring, and follow-up:
  - Prioritize drugs for discontinuation and plan the order of discontinuation. Medications which are causing (or have high risk of causing) harm, and those which are of greatest concern to the patient, should be stopped first.

    Stopping one drug at a time is usually recommended to encourage patient willingness and enable appropriate monitoring. However, in the case of suspected adverse drug reactions or minimal risk of an adverse drug withdrawal event (ADWE) such as with vitamins, two or more drugs can be withdrawn simultaneously.
  - Develop a deprescribing plan with the patient and other relevant health care professionals. A few key points to discuss with the patient include:
    - Deprescribing should be considered a trial, especially when it is unclear whether the medication is still providing a benefit (such as PPIs for GERD). Knowledge that deprescribing is a trial rather than an irreversible decision can reassure patients.
    - Tapering is recommended where there is risk of adverse drug withdrawal reactions (ADWRs) or if there is concern that the underlying condition will return or worsen. In this setting, tapering can help identify the lowest effective dose, minimize return of symptoms if they occur, and encourage patient willingness to have a medication deprescribed.

  - Patients (and their caregivers or family) should be aware of what to self-monitor for, what to do if they notice a change, and the need to attend a follow-up appointment for monitoring with a health care professional. Frequency of monitoring should be clear as well as whether it can be done via telehealth, electronic messaging, in person, or a combination of these.

- Implement and monitor the deprescribing plan.

- Document the plan for deprescribing and any outcomes (eg, if a medication is restarted, note why and whether deprescribing should be re-attempted).

- Ensure ongoing communication with the patient (and their caregivers or family) and relevant health care professionals.

Tapering doses — For most medications, there is limited evidence and few guideline recommendations on how to conduct withdrawal. If there is any doubt about whether a medication can be stopped abruptly, it is safer to taper the dose over weeks to months.

Tapering can reduce the chance of ADWEs, facilitate identification of the lowest effective dose in patients who are unable to stop a drug completely, and support long-term drug cessation by increasing patient comfort and willingness to try deprescribing [71]. (See 'Deprescribing specific medications' below.)

In our experience, a general rule of thumb is to reduce the dose by 50 percent every two to four weeks, with monitoring at each dose reduction and at two to four weeks after cessation. Stepping down through available dose formulations is another approach. However, some guidance recommends a slower rate of taper, especially with psychotropic medications (eg 5 to 25 percent every two to four weeks). For many psychotropics, a hyperbolic tapering approach that accounts for nonlinear relationships between drug dose and receptor occupancy has been advocated. This involves applying a percentage dose reduction to each new step-down dose rather than to the original dose (for example, a 50 percent hyperbolic taper of a 20 mg daily dose would involve stepping down from 20 mg, to 10 mg, to 5 mg, to 2.5 mg, and so on).

However, in many situations there is little evidence to guide optimal tapering regimens, and which patient characteristics are associated with the greatest risk of ADWRs. Factors that may indicate the need for a slower taper include long duration of use, high doses, previous failed tapers and short half-life [78-81].

Preventing adverse drug withdrawal events — An adverse drug withdrawal event (ADWE) is a clinically significant negative outcome caused by drug discontinuation [82,83]. ADWEs may encompass unwelcome symptoms including the return of the medical condition that the drug had been used to treat, or a physiological withdrawal reaction (eg, rebound phenomena). The latter, a subset of the broader category of ADWEs, is often referred to as an adverse drug withdrawal reaction (ADWR) [81]. These typically occur when a medication is withdrawn too quickly (eg, stopped abruptly).

A number of drug classes have been associated with ADWRs, for example, beta blockers, corticosteroids, and benzodiazepines [84]. A list of drug classes associated with ADWRs and their associated symptoms is provided in the table (table 2).

A slow taper can in most, but not all, cases prevent ADWRs [85-93] (see 'Deprescribing specific medications' below). When deprescribing a medication that is associated with an ADWR, closer monitoring, including informing the patient what to self-monitor for, is important, particularly in the first few weeks following discontinuation [94-97].

Medications that can result in adverse drug withdrawal reactions if stopped abruptly*[1-6]

Medication	Symptoms that can result from abrupt discontinuation
Acetylcholinesterase inhibitors	Abrupt cognitive decline, aggression, agitation, hallucinations, reduced consciousness
Antiseizure medications	Anxiety, depression, seizures
Antidepressants	Akathisia, anxiety, chills, gastrointestinal distress, headache, insomnia, irritability, malaise, myalgia
Antiparkinsonian agents	Hypotension, psychosis, pulmonary embolism, rigidity, tremor
Antipsychotics	Dyskinesias, insomnia, nausea, restlessness
Baclofen	Agitation, anxiety, confusion, depression, hallucinations, hypertonia, insomnia, mania, nightmares, paranoia, seizures
Benzodiazepine receptor agonists	Agitation, anxiety, confusion, delirium, insomnia, seizures
Beta blockers	Angina pectoris, anxiety, myocardial infarction, severe hypertension, tachycardia
Clonidine	Agitation, headache, palpitations, severe hypertension
Corticosteroids	Anorexia, hypotension, nausea, weakness
Opioids	Abdominal cramping, anxiety, chills, diaphoresis, diarrhea, insomnia, restlessness
Proton pump inhibitors	Gastric upset, heartburn (due to rebound hyperacidemia)

Medications that can result in adverse drug withdrawal reactions if stopped abruptly*[1-6]

Table 2 - larger image below

In certain cases, the biologic and clinical outcomes of stopping a medication used over the long term may not simply be the reverse of starting it, with potential contributing factors including up- or down-regulation of biologic pathways and the "weeding out" of medication-intolerant patients early in a treatment course. For example, preclinical studies of statins suggest a potential rebound effect in biologic pathways after these drugs are stopped, although the clinical significance of this remains unclear [98]. In the case of aspirin used for primary prevention, the ASPREE trial found that chronic users who discontinued this medication had a nonsignificant statistical trend toward worse clinical outcomes than people who continued it, an opposite result compared with people newly starting this medication [99]. For this reason, the American Geriatrics Society Beers Criteria draws a distinction between starting aspirin for primary prevention (recommendation to *avoid*) [65] and discontinuing the drug among chronic users using it for primary prevention (recommendation to *consider deprescribing*).

Practical strategies — Some practical tips for successful deprescribing include:

- Set aside separate visit(s) to focus on medication review and deprescribing. Reimbursement/billing codes can include the diagnosis that the medication is intended to treat or general wellness/annual health visits.

- Use the expertise and services of pharmacists, nurses, and other health care professionals, either to lead the process, to complete time intensive tasks, or as part of a multidisciplinary team [100-105]. Programs that engage such professionals include Medicare Part D

Medication Therapy Management (MTM) programs in the United States [106] and Home Medicines Reviews in Australia [107].

- Start the process; deprescribing doesn't need to be conducted all in one visit.
- Use patient educational materials (PEMs) when possible. (See 'Communication with patients and families' below.)

References that describe other stepped approaches to deprescribing are also available [70-72,108-111].

DEPRESCRIBING SPECIFIC MEDICATIONS

Information regarding deprescribing of certain commonly prescribed medication classes is provided below. Online resources to assist with deprescribing of particular medications are available. (See 'Resources to support deprescribing' below.)

Benzodiazepines and benzodiazepine receptor agonists — Because both psychological and physiologic dependence can occur with these medications, closely engaging patients, offering alternate nonpharmacologic or pharmacologic therapies, and obtaining buy-in before and during discontinuation attempts is essential [85,112,113]. (See "Overview of the treatment of insomnia in adults".) A guideline on benzodiazepine tapering published in 2025 provides a useful overview of strategies [114].

- Very slow tapers may be needed, particularly when these drugs have been used at high doses or longer durations (eg, more than three months). Recent guidelines recommend a 5 to 10 percent dose reduction every two to four weeks and typically should not exceed more than a 25 percent dose reduction every two weeks [114]. A slow approach is also often necessary at the end of the taper, for example, if an initial 25 percent reduction is used, using 12.5 percent reductions, to help make the final transition to abstinence [85,86]. Thus, a full taper can take months to years, especially for people who have used high doses for an extended period. However, it should be noted that there is very limited evidence to guide optimal tapers with some studies employing much shorter tapers reporting minimal withdrawal reactions [115]. More rapid tapers can be feasible in some people, in particular those taking low doses or with use for a relatively short duration (eg, less than there months).
- Use of Cognitive Behavioral Therapy for Insomnia (CBT-I) while tapering can increase the likelihood of successful cessation. health care. Although there can be cost and access barriers to therapist-delivered CBT-I [116,117], online, app-based, and other self-directed programs can also be effective. For example, a package of two CBTI-informed booklets and a companion website from Canada's Sleepwell initiative increased the success of

benzodiazepine or benzodiazepine receptor agonist tapering [118]. Other free online and app-based CBTI programs are available through the United States Department of Veterans Affairs, and other sources [67,119].

- A useful tapering schedule and patient education materials (PEMs) used in the EMPOWER and D-PRESCRIBE trials are also available [5,120].

Withdrawal symptoms (eg, insomnia, anxiety, irritability, gastrointestinal symptoms) can occur, particularly once doses have been reduced to approximately 25 percent of the original dose among people with long-term, high-dose use [85].

- If such symptoms appear, patients should be reassured that they are usually mild and subside in days to several weeks.
- Maintain the current dose for one to two weeks and then resume the taper [86] with consideration given to slowing the rate of the taper [114].

For patients who have had unsuccessful attempts at deprescribing, understanding reasons for lack of success (eg, overly rapid tapering) can be used to formulate new discontinuation plans. With proper intervention and support, up to 60 to 80 percent of benzodiazepine users have been able to stop using these medications [85]. Of note, stopping benzodiazepines and benzodiazepine receptor agonists does not guarantee improved outcomes, and observational studies have found neutral or even paradoxically increased risk of adverse outcomes in long-term benzodiazepine users who stopped versus continued [121,122]. While unmeasured confounding and/or poor tapering approaches may explain these findings, a thoughtful approach to patient selection and deprescribing strategy, including avoiding overly rapid tapers and ensuring that patients do not substitute benzodiazepines with even more harmful ways of managing their insomnia or anxiety, is warranted [114].

Proton pump inhibitors — We have a low threshold for targeting proton pump inhibitors (PPIs) for deprescribing, as they are commonly overused and often continued indefinitely rather than for the discrete duration appropriate for the indication [123]. Harms of treatment include increased risk of *Clostridioides difficile* colitis, possibly hip fractures, and impaired vitamin B12 absorption.

Before deprescribing, ensure that the patient does not have a compelling reason for ongoing treatment, such as Barrett's esophagus, chronic nonsteroidal antiinflammatory (NSAID) use with bleeding risk, severe esophagitis, history of bleeding upper gastrointestinal ulceration, and in some cases older age with chronic anticoagulant or corticosteroid use [123-125]. If the patient does not have a condition which necessitates long term use, and they have used the PPI for more than four to eight weeks, then deprescribing is generally recommended.

Although evidence about optimal deprescribing strategies is limited, gradual dose reduction may be more effective than abrupt discontinuation given risk of rebound hyperacidemia with the latter [87,88,125-129]. One reasonable approach is as follows:

- Halve the dose for two to four weeks, then stop; for people already on the lowest dose, alternate-day therapy may be useful [88].
- Monitor at 4 and 12 weeks for symptom recurrence (eg, heartburn, dyspepsia, regurgitation, or among non-verbal people anorexia or agitation).
- Occasional symptoms may be managed with on-demand therapy (antacids, H2 receptor antagonists, or PPIs) or with daily H2 receptor antagonist use. On-demand therapy with PPIs has been recommended as a form of minimizing total use and is associated with high patient satisfaction [88], although the drug's mechanism of action may not be conductive to irregular use [130]. H2 receptor antagonists may also be substituted for PPIs, although with higher risk of symptom return than with ongoing use of lower-dose PPIs [123].
- If symptoms persist and interfere with normal activity, consider testing for *Helicobacter pylori* and returning to the lowest effective PPI dose. *H. pylori* treatment is discussed elsewhere. (See "Treatment of *Helicobacter pylori* infection in adults".)
- Most patients who have an indication for chronic PPI use and are treated with twice-daily dosing can be considered for reduction to once-daily use [125].

Antipsychotics for behavioral and psychological symptoms of dementia or for insomnia — Since antipsychotics confer increased risk of mortality in older adults with dementia, trials of discontinuation are warranted for patients whose behavioral and psychological symptoms have stabilized or who did not improve on antipsychotic therapy (while recognizing that such symptoms often fluctuate or change over time even without intervention) [9,89,131]. Withdrawal of antipsychotics in people with dementia may not significantly affect behavioral and psychological symptoms [132,133]. Treatment of neuropsychiatric symptoms in patients with dementia, including the use of antipsychotic medications, is discussed elsewhere. (See "Management of neuropsychiatric symptoms of dementia".)

Although there is little clinical evidence to support one particular tapering scheme [134], a recommended strategy is as follows:

- Reduce the dose by 25 percent, then 50 percent, then 75 percent every one to two weeks, then stop [89,90,131].
- At each step, there should be close follow-up for potential adverse drug withdrawal events (ADWEs) such as worsening psychosis, aggression, or hallucinations.
- Behavioral and environmental strategies for symptom control, the mainstay of treatment, should be continued throughout. (See "Treatment of Alzheimer disease", section on

'Nonpharmacologic therapy and supportive care'.)

- Relapse of symptoms may be managed by nonpharmacologic means or restarting drug therapy at the lowest possible dose with regular review and consideration for a retrial of deprescribing.

In the setting of insomnia treatment with low-dose antipsychotics, tapering may not be necessary, and persistent symptoms should ideally be managed with behavioral approaches [90]. Expert consultation is advised before considering deprescribing for other indications, such as schizophrenia or adjunctive treatment for major depression.

Glucose-lowering medications — Deprescribing glucose-lowering medications may be warranted in patients for whom potential risks are likely to exceed benefits [135,136]. Such patients include those with multiple chronic conditions, renal impairment, dementia, limited life expectancy, history of hypoglycemia, impaired ability to sense or respond to hypoglycemia (eg, concurrent use of beta blockers, cognitive impairment), or use of agents most likely to precipitate hypoglycemia (insulins, sulfonylureas).

The "4S" pathway, developed by an international expert panel, provides a useful four-step roadmap for addressing deprescribing of glucose-lowering medications in older adults [137]:

- Step 1 involves seeking triggers that might prompt considering of deintensifying glycemic control; these can include medical events such as falls, life events such as a change in living situation or loss of a care partner, worsening physical or cognitive status, and medication-related red flags such as complex regimens, suspected or confirmed hypoglycemia, or other medication adverse effects.

- Step 2 involves shared decision-making. This is essential, since patients who have been told for decades about the importance of tight glycemic control may have difficulty accepting a new message that tight control can now cause them more harm than good, or they may hold values and understandings that run counter to guidelines [138-141]. It is also critical to communicate and coordinate care with other clinicians involved, especially any endocrine specialists.

- Step 3 comprises setting goals, which can include hemoglobin A1c and/or serum blood glucose targets as well as minimizing hypoglycemia. Common glycemic management goals, which can vary based on a variety of factors, are reviewed elsewhere including for older adults. (See "Overview of general medical care in nonpregnant adults with diabetes mellitus", section on 'Glycemic management' and "Treatment of type 2 diabetes mellitus in the older patient", section on 'Goals'.)

- Step 4 involves aligning the treatment regimen to reflect these goals, maximize simplicity where appropriate, and reduce harm.

When undertaking such realignment, it is typically preferable to focus first on discontinuing or reducing the dose of agents at highest risk of causing hypoglycemia (eg, insulins, sulfonylureas). While there is little evidence to support a specific approach to deprescribing [142,143], reasonable strategies include gradual dose lowering every one to four weeks, switching to a safer medication, or, when severe hyperglycemia is not anticipated, abrupt cessation. For patients switching from complex to simpler insulin regimens, the American Diabetes Association guidelines offer an algorithm [144], and the "4S" expert panel also provides useful algorithms for both insulin and oral regimens. Finally, for people taking an SGLT2 inhibitor or GLP1 receptor agonist, consider if the drug may confer benefit for a nondiabetes indication (eg, heart failure, chronic kidney disease progression) even if it is no longer necessary for glycemic control [137].

After a change is made, instruct the patient to monitor daily for one to two weeks for signs of severe hyperglycemia; these can be assessed by symptoms such as fatigue or excessive thirst or urination and typically do not require regular fingerstick blood glucose measurements for patients not already taking such measurements. Changes in hemoglobin A1c levels may not occur for several months [142,145]. Limited evidence suggests that deprescribing often does not worsen glycemic control [143,146,147].

Cholinesterase inhibitors and memantine — Decisions about deprescribing cholinesterase inhibitors and memantine can be very challenging due to uncertainty about whether these medications are providing benefit to any given patient. Deprescribing should be considered in patients who do not have an evidence-supported indication (eg, mild cognitive impairment), in patients who have been taking the medication long term with substantial interval worsening of their cognition and/or function or with lack of benefit from treatment, and in patients with severe and end-stage dementia [148,149]. Other reasons to pursue deprescribing include adverse effects (eg, syncope, gastrointestinal distress, or weight loss with cholinesterase inhibitors), nonadherence and possibly severe agitation (given some evidence that cholinesterase inhibitors can worsen this symptom, although limited and inconsistent studies suggest that discontinuation may result in worse neuropsychiatric symptoms in the short to medium term) [149-151]. Randomized trials have found that people who discontinue cholinesterase inhibitors have, on average, a likely clinically meaningful worsening of cognitive function and possibly neuropsychiatric symptoms compared with those continuing the medication, although with no differences between groups in global assessment of change or quality of life [148,150,152]. These trials may not be generalizable to all people with dementia, particularly those who are older, are in residential care facilities, or have multiple comorbidities. Decisions about deprescribing of cholinesterase inhibitors and memantine should be conducted via shared decision making with the patient, caregivers, and family members.

The use of cholinesterase inhibitors, including decisions to discontinue therapy, is discussed in detail separately. (See "Cholinesterase inhibitors in the treatment of dementia", section on 'Duration of therapy'.)

Once the decision to deprescribe has been made, there is little information to inform stopping strategies. However, we suggest the following:

- Tapering of dose and close monitoring for cognitive, functional, and neuropsychiatric symptoms are strongly advisable, as case reports suggest that severe adverse drug withdrawal reactions (ADWRs) can occur from abrupt discontinuation of cholinesterase inhibitors and memantine [91,153,154]. Tapering should involve halving the dose every four weeks to the lowest available dose, then stopping; for medications where the dose formulation cannot easily be halved, stepwise reduction through the available dose formulations is reasonable [91].

- A reasonable follow-up monitoring interval is four weeks after each dose reduction, or one to two weeks for patients at higher risk of ADWRs [91,148,149].

  - If substantial neuropsychiatric changes occur in the first week after a dose reduction or discontinuation, (eg, agitation, aggression, hallucinations, or reduced consciousness) this may indicate an ADWR, and the previous dose should be promptly restarted. After stabilization, it is reasonable to attempt deprescribing again with a more gradual taper.

  - If cognitive, behavioral, or psychological symptoms emerge in roughly two to six weeks, this may indicate that the drug was providing benefit, and restarting at the previous dose should be considered (unless there is an apparent other cause of these symptoms, for example urinary tract infection or initiation of another psychotropic medication).

  - If such symptoms emerge between 6 and 12 weeks, this may indicate either the above or progression of the underlying condition [91,148,155].

Antidepressants — Indications for discontinuation of antidepressants are discussed in detail elsewhere (See "Antidepressant discontinuation syndrome and discontinuing antidepressants in adults".)

Withdrawal symptoms from antidepressant discontinuation are common and may include insomnia, increased anxiety, dizziness, nausea, and flu-like symptoms [156,157]. Although withdrawal symptoms often resolve within one to two weeks, they can persist for weeks to months [156,158], and care should be taken to distinguish between withdrawal symptoms and relapse of the underlying condition the antidepressants were used to treat [92,159]. The incidence of withdrawal symptoms can be reduced by tapering doses gradually over a period of weeks [92,93,158]. Some have argued for even longer tapers, with progressively smaller dose decrements over several months [160]. However, there is limited evidence as to which dose-reduction schedules work best [92,93,161].

Antihypertensive medications — Blood pressure targets for older adults are controversial and continue to evolve [162,163]. Discontinuation of antihypertensives may be considered when blood pressure is below targets, as well as in the setting of adverse drug events, drug-drug interactions, end-of-life, and patient preference [164-166].

In a systematic review of discontinuation of antihypertensives used for hypertension or primary prevention in older adults, rates of adverse events attributed to drug withdrawal remained low, and some metabolic derangements improved once therapy was withdrawn. However, a substantial proportion of patients were restarted on therapy due to elevated blood pressures [167].

- **Frailty** – There is no consensus on the role of deprescribing antihypertensive medications among older adults with frailty, including those residing in nursing homes. Several studies suggest that a judicious approach to deprescribing in such patients can have neutral effects on mortality and cardiovascular events and potential small benefits on other outcomes important to older adults [168-172]. In a multicenter trial among 1048 nursing home residents 80 years of age or older (65 percent with moderate or severe frailty) who were taking two or more antihypertensive drugs and had a systolic blood pressure below 130 mm Hg, those randomized to receive a strategy to deintensify antihypertensive treatment had similar rates of all-cause mortality (adjusted hazard ratio [aHR] 1.02, 95% CI 0.86 - 1.21) and major adverse cardiovascular events (aHR 1.15, 95% CI 0.84-1.56) after a median follow-up of 38 months compared with those receiving usual care [173]. A separate series of observational studies of antihypertensive deprescribing in nursing home residents found similarly null effects on cardiovascular outcomes and potential small benefits on cognition and on functional status (among those without dementia) [165,171,172].

Some antihypertensive medications should never be stopped abruptly due to the risk of potentially fatal withdrawal syndromes. These include short-acting beta blockers (eg, propranolol) and alpha-2 agonists (eg, clonidine, guanfacine, guanabenz). Safe discontinuation of these medications requires slow tapering with close monitoring. Other antihypertensives, including in most cases ACE inhibitors, thiazide diuretics, and calcium channel blockers, do not carry this risk of physiologic rebound and can be stopped without tapering. However, we encourage home or clinic-based follow-up for all antihypertensive tapering or discontinuation to evaluate response to medication withdrawal. Deprescribing antihypertensives and their associated withdrawal syndromes are discussed in further detail separately. (See "Tapering and discontinuing antihypertensive medications".)

SHARED DECISION MAKING

Communication with patients and families — Effective communication with patients about goals and preferences and experiences with their medications is essential to identify medications which are suitable for deprescribing (table 3) (see 'Approach to deprescribing' above). It also helps to engage the patient in the deprescribing process, including using patient education materials (PEMs) as needed, to prime them for further discussion [174,175].

Points to consider when communicating with patients and families or other decision makers include:

- Determine their preferences for involvement in the decision-making process. Some patients may prefer to defer to their health care professional for decision making; this does not negate the need for good communication and shared decision making, but it can help clinicians tailor the discussion to the individual [176-180].
- Discuss patient goals and preferences and elicit patient's experiences with their medications in order to identify medications suitable for deprescribing. Understanding their attitudes toward deprescribing and tailoring deprescribing recommendations to "what matters" for the patient provides a patient-centered approach and maximizes opportunities for buy-in [73,181,182].
- Discuss the "why" (why should the medicine be stopped?) and "how" (how will the withdrawal and monitoring be done?) of deprescribing [138,183-186]
- Discussion of the risk of side effects as a rationale for deprescribing is generally preferred by older adults [187]. Combining a discussion of the risk of side effects or potential for harm with rationale about a lack of benefit may optimize patient acceptance of deprescribing [188,189].
- Frame deprescribing as a trial, with reassurance that the medication(s) can be restarted if necessary.
- Frame deprescribing as "not giving up on you" or taking something of value away, but as optimizing care. Discuss what alternative strategies are being taken to manage the symptom or disease the deprescribed medication was intended to treat.
- Elicit and address concerns and fears about deprescribing:
  - Patients may be reluctant to stop a medication if they think it is still necessary and may experience cognitive dissonance from being told to stop a medicine that for years they were urged to take or was described as "lifelong treatment." Other concerns may include fear of their condition returning, fear of a withdrawal reaction, or a nonspecific fear of change [138,185,186]. Such emotions about deprescribing can easily overtake logic. Unless there is imminent risk of harm from a medication, addressing emotions about medication use before implementing a deprescribing plan can maintain therapeutic

alignment between the clinician and patient and increase chances of long-term success from deprescribing attempts.

- If the patient is against medication withdrawal, the reasons for this should be explored (eg, fear or belief the medication is necessary). If they are well-informed about the likely risks and benefits and place a greater value on the potential benefit than the likely harm of continuation, then deprescribing may not be appropriate (unless there is a clear and present risk of harm) [190]. However, the conversation can be revisited at regular intervals, particularly when there are changes in the patient's condition or new evidence to inform deprescribing.

- The use of PEMs may aid in communication and shared decision making around deprescribing [175]. PEMs are available about deprescribing in general as well as for specific medications such as antidepressants, psychotropics, proton pump inhibitors (PPIs) and cholinesterase inhibitors [191-194]. Although there is scarce information on the development, implementation, and outcomes of use of most tools, PEMS can be a strong trigger for deprescribing [175]. For example, in a trial of older adults, the EMPOWER brochure, a PEM designed to induce cognitive dissonance about sedative-hypnotic use and provide user-friendly guidance on how to deprescribe, led to a 27 percent reduction in use of benzodiazepines [120].

Communication with other health care professionals — Clinicians may be hesitant to stop a medication that was started by a specialist or while the patient was in the hospital. This may be because they do not feel that they are responsible for that medication, because of professional hierarchy, or because they do not wish to affect the patient's relationship with this other professional [139,142]. However, these clinician barriers can be addressed.

Direct communication between health care professionals (such as specialists and primary care clinicians) when deprescribing is being considered can be invaluable to resolve uncertainties, prevent conflicting instructions to patients, and ensure alignment of the therapeutic plan [195-197]. However, if communication is not forthcoming, clinicians should not hesitate to take the initiative in stopping a medication that is clearly inappropriate or causing more harm than benefit.

Empowering patients to advocate for clear instructions when a medication is started or stopped can enhance communication between all parties. The patient may be primed to act as a conduit between the primary care clinician and specialist, for example "next time you see your cardiologist, make sure to tell them about this symptom you are having, and ask if it's safe to stop this medication."

When a non-prescribing clinician (such as a pharmacist) is making recommendations to a primary care clinician, having clear information on why the medication should be deprescribed (with supporting evidence where possible) as well as how to conduct deprescribing will likely increase the uptake of the recommendations. Templates for communication of deprescribing recommendations between pharmacists and clinicians ("pharmaceutical opinions") have been developed and trialed in Canada with favorable results [198,199].

COMMON BARRIERS TO DEPRESCRIBING AND STRATEGIES FOR OVERCOMING THEM

Patient and/or family reluctance — Patients and/or families may be reluctant to engage in deprescribing. However, clinicians may perceive reluctance, based on past experience, even if patients have not specifically expressed such concerns [100,139,200].

Most strategies for overcoming real or perceived reluctance relate to good communication. (See 'Communication with patients and families' above.)

It is often helpful to engage with patients and/or family early in the process, to communicate the "why" and "how" of deprescribing, and to address specific fears (table 3).

While deprescribing may not be familiar to patients, studies have shown that approximately 90 percent of older adults and caregivers say that they would be willing to stop one or more of their medicines if their doctor said it was possible [183,184,201].

The use of existing patient educational materials (PEMs) may be helpful [175].

Lack of evidence — There is a relative lack of evidence and guidelines to inform deprescribing (compared with initiation of medications). This, coupled with uncertainly over whether the medication is beneficial to the individual, can mean that there is limited objective impetus to deprescribe and fear about the outcomes of deprescribing [139,190,196,202]. Strategies to reduce uncertainty include:

- For medications used to treat symptoms or whose effectiveness can be measured by a biomarker (eg, blood pressure), considering a deprescribing trial with monitoring for symptoms and/or worsening or return of condition.
- Weighing the lack of evidence for the benefits of continuing a medicine (particularly among older adults with polypharmacy and multimorbidity), not only the lack of evidence for deprescribing.
- Prioritize a patient-centered approach, focusing on the patients' preferences with knowledge of their unique circumstances [202].

Limited time — Deprescribing is a complex process, which in the majority of cases requires considerable time to undertake. As such, the ability to deprescribe within a single clinical encounter may be limited [139,203]. Health care professionals can overcome time limitations by utilizing practical strategies. (See 'Practical strategies' above.)

Care shared among multiple providers — When medications are prescribed or recommended by another clinician such as a specialist, primary care clinicians may feel like it's not their role to manage the medications (devolving of responsibility) or may not wish to make changes due to professional hierarchies [202-206]. Additionally, they may be concerned about damaging the patient's relationship with this other professional by providing contradictory advice. However, it is helpful to not assume that specialists are resistant to deprescribing, and to communicate clearly. (See 'Communication with other health care professionals' above.)

Unintended restarts — In many electronic health records, removing a medication from a patient's electronic medication list does not result in a discontinuation order being sent to that patient's pharmacy. As a result, that pharmacy may continue to automatically refill that medication or dispense it if requested by the patient. Similarly, in-hospital medication discontinuations may not be communicated to primary care providers, community pharmacies or the patients themselves [207,208]. To avoid this, it can be helpful to contact (eg, by phone) the patient's pharmacy to ensure that the medication is discontinued. An information technology solution for transmitting this information automatically is employed in some health systems but is not yet widespread [209].

Challenges in recognizing problematic medications — A lack of recognition of inappropriate medication use in individual patients has been identified as a barrier to deprescribing [139]. This may be addressed through:

- Using electronic prescribing support and audit and feedback services where available to enable proactive deprescribing.
- Realigning expectations of benefits and harms of medication use (both patients and clinicians tend to overestimate the benefits of medication use and underestimate the harms) [210,211].
- Considering adverse drug reactions in the differential diagnosis of new symptoms or a change in condition [212,213].

Medical culture and clinical inertia — Medical culture has been highlighted as a barrier to deprescribing, including a historically clinician-centric culture where prescribing is a central part of professional identity [206,214-216]. Additionally, starting a medication is familiar and considered a positive action (ie, doing something to help the patient), while deprescribing is less familiar and may be considered a lower priority or as withdrawing care. Clinical inertia (continuation along a path of treatment without re-evaluation or staying with the "status quo") is also common in medical culture and can discourage deprescribing [196,203,217].

Strategies to combat clinical inertia and cultural norms and increase the normality of deprescribing include:

- Equally considering the benefits and harms of continuation against the benefits and harms of discontinuation [190].
- Attending deprescribing-related continuing education opportunities and advocating for greater undergraduate teaching of deprescribing.
- Discussing deprescribing activities with colleagues, including success stories [202].
- Integrating deprescribing into undergraduate medical, pharmacy, and other health care professional education [218].
- Inclusion of deprescribing recommendations in clinical practice guidelines [219,220].

SPECIAL POPULATIONS AND SETTINGS

Hospitalized patients — Hospitalization is a potentially opportune time for deprescribing. There is often a complete medication history and reconciliation performed on admission, access to a multidisciplinary team (including specialists), clear impetus to address medications that may have contributed to the reason for admission (eg, fall-risk increasing medications in an older adult admitted for fall-related injury), and the ability for close short-term monitoring. However, there are also limitations: duration of admission may be considered too short to complete a deprescribing process, working patterns may limit capacity to support deprescribing, deprescribing of medications unrelated to the reason for admission may be a lower priority, and inpatient clinicians may not be aware of the complete history and context of a patient's medication use [200,221]. Communication with the primary care clinician about potential deprescribing targets is generally advisable before changing long-term medications [197,222-225]; if such communication is not forthcoming, caution should be taken with changing treatment for chronic conditions not related to the reason for hospitalization, although it is reasonable to make changes where a medication is clearly harmful or inappropriate. Trials of interventions to support medication review and deprescribing in hospitals and post-acute care settings have successfully stopped medications and systematic reviews have found that such interventions confer a modest reduction in short-term readmission rates, but without clear impacts on other clinical outcomes such as adverse drug events [226-229]. As in other settings, this highlights the complexities of deprescribing in this population, and stopping medications without any change in clinical outcomes may be seen as a success [28,229].

Medications are commonly added during hospitalization and continued after discharge even when no longer needed, for example, medications used for stress ulcer prophylaxis and those used to treat constipation, insomnia, or elevated blood sugar or blood pressure that occur during inpatient stays. These medications may be continued indefinitely unless the watchful clinician identifies and stops them. Post-hospitalization visits, for example to primary care clinicians, should thus be used to review all medications changed during the hospital stay and stop those which are no longer needed.

People nearing the end of life — As life expectancy shortens, people often prioritize controlling symptoms over life-preserving therapy and there is less opportunity to benefit from therapies designed to prevent disease over the long term, for example, anticoagulants, cholesterol-lowering therapies, and anti-resorptive therapies for osteoporosis [54-56,230,231] (see 'Patient characteristics' above). Several research groups have compiled lists of medications that are often inappropriate near the end of life [54,232-235]. Such medications can thus often be useful to deprescribe, although good communication remains paramount. Conversely, some medications such as benzodiazepines, which may otherwise be frequent targets of deprescribing efforts in older adults, can be useful in palliative care. The attitudes held by patients nearing the end of life and their caregivers towards deprescribing can vary and, as with the general population, conversations need to be tailored to the individual [236].

Patients with dementia — Older adults with dementia are commonly prescribed medications that can negatively impact their cognitive and physical health, including antipsychotics, anticholinergic medications, and psychotropic polypharmacy. While appropriate for some people, these are often good candidates for deprescribing. Other considerations that should inform deprescribing decisions include treatment burden and patient resistance to taking medications (including the role of caregivers in these dynamics), the availability of caregiver supports to help administer medications and promote their safe use, and goals of care [237].

Goals of care often change as people develop dementia and progress to more advanced states of this condition. While preferences can vary substantially in people with milder forms of dementia, there is general consensus that deprescribing of preventive and nonsymptom-control medications is appropriate in those with advanced dementia given limited life expectancy. Several groups have developed consensus lists of medications to avoid in people with advanced dementia [231,238,239]. While there is substantial heterogeneity between these lists, drugs which most of these groups recommend avoiding include anticoagulants, antiplatelets, alpha-blockers, anti-arrhythmic medications, antihypertensives, lipid-lowering therapies, and several genitourinary and sex-hormone medications including antiandrogens, antiestrogens, sex hormones, and hormone antagonists [231]. However, as in other situations it is important to achieve agreement and maintain good communication with patients and their caregivers before stopping such medications. There may be additional barriers to deprescribing in this population due to changing care goals, uncertain disease trajectory, inadequate guidelines and the involvement of caregivers (and their eventual progression to being a substitute decision maker) [59,240]. (See "Care of patients with advanced dementia", section on 'Discontinuing chronic medications'.)

Children — There is little published literature on deprescribing in children and adolescents. While polypharmacy is commonly associated with ageing and older adults, it can occur at any age. Possibly because of the lack of evidence-based guidelines for children, a culture of "trialing" medications often occurs. While deprescribing may be more complex in children due to a lack of evidence for both prescribing and deprescribing, it may be easier to establish indications due to the use of electronic health records and close involvement of parents/caregivers [241]. There is a need for further investigation and development of interventions and support tools in this area, particularly in pediatric psychiatry [242,243], or in areas where new medications may reduce the need for polypharmacy such as cystic fibrosis [244].

RESOURCES TO SUPPORT DEPRESCRIBING

There are a large variety of resources and tools that have been developed to support health care professionals to deprescribe, such as generic frameworks and drug-specific deprescribing guidelines. These tools have been developed using different methods such as expert consensus, structured and unstructured literature reviews, external review, or following internationally recognized standards (such as guideline development) [245].

Evidence-based guidelines for deprescribing have been developed by a Canadian consortium for an expanding number of medication classes, with easy-to-use algorithms and other resources available for online review and download. Their methods have also been applied to develop an evidence-based guideline for deprescribing opioids [246], and there are ongoing efforts to expand deprescribing recommendations in disease-focused clinical practice guidelines [220]. Trial-tested patient handouts that are designed to address the educational, psychological, and behavioral needs of the patient can also be helpful.

Other online resources for deprescribing of specific medications are also available from the NSW Therapeutic Advisory Group, the Canadian Deprescribing Network, Primary Health Tasmania, the Australian Deprescribing Network, and the US Deprescribing Research Network. Additional information on patient education materials are presented above. (See 'Communication with patients and families' above.)

Individual health systems may have additional resources such as pharmacist consultation services or deprescribing clinics such as the IMPROVE polypharmacy project and interventions developed through the VA Center for Medication Safety and the United States Department of Veterans Affairs' VIONE initiative [247,248]. Independent consultant pharmacists can be found through the American Society of Consultant Pharmacists website [249] and can provide valuable services.

Decision support tools and deprescribing reminders integrated into electronic medical records are increasingly being evaluated. In the long term care setting, an electronic decision support paired with routine medication reviews improved deprescribing, including potentially inappropriate medications [250]. An outpatient intervention of sequential EHR notifications during patient visits, prompting providers to initiate deprescribing discussions and reminding providers to deprescribe at the next visit, increased deprescribing rates by 10.4 percent compared with usual care [251]. Similarly, integration of the Drug Burden Index (DBI: a clinical risk assessment tool that represents total exposure to sedative and anticholinergic medications) into hospital medical records, paired with a 'polypharmacy steward' has been shown to facilitate deprescribing in older inpatients [252,253].

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topic (see "Patient education: Medication safety (The Basics)")

SUMMARY AND RECOMMENDATIONS

- **Definition** – The term "deprescribing" refers to a process of medication withdrawal, supervised by a health care professional, with the goal of managing polypharmacy and improving outcomes [1]. This can encompass efforts to comprehensively review a patient's medication list and systematically discontinue or reduce the dose of all medications with an unfavorable balance of benefits and harms, as well as efforts focused on specific high-risk medications. (See 'Definition' above.)

- **Goals of deprescribing** – Common goals for deprescribing include reducing overall medication burden, reducing the risk of specific geriatric syndromes such as falls and cognitive impairment, and improving global health outcomes such as hospitalization and death. (See 'Goals of deprescribing' above.)

- **Patient characteristics** – Patient characteristics which are good targets for deprescribing efforts include polypharmacy, multimorbidity, kidney function impairment, transitions of care, medication nonadherence, limited life expectancy, older age, frailty, and dementia. (See 'Patient characteristics' above.)

- **Medications** – Commonly overused and high-risk medications are good targets for deprescribing (table 1). (See 'Medications' above.)

  - Specific information regarding deprescribing of certain medication classes is available, including benzodiazepines and benzodiazepine receptor agonists, proton pump inhibitors (PPIs), antipsychotics, glucose-lowering medications, cholinesterase inhibitors and memantine, antidepressants, and antihypertensive medications. (See 'Deprescribing specific medications' above.)

- **Stepwise approach** – Deprescribing is best accomplished in a stepwise approach which includes engaging the patient and gathering information, identifying and deciding on medications to deprescribe, and implementing a deprescribing plan with monitoring and follow-up (table 3). (See 'Stepwise approach' above.)

- **Tapering** – Tapering is a good strategy for many medications. It can reduce the chance of adverse drug withdrawal reactions (ADWRs), facilitate identification of the lowest effective dose in patients who are unable to stop a drug completely, and support long-term drug cessation by increasing patient comfort and willingness to try deprescribing. Certain medications are more likely to cause adverse drug withdrawal reactions (ADWRs) if stopped abruptly (table 2). (See 'Tapering doses' above.)

- **Shared decision making** – Shared decision making is essential to successful deprescribing and should include alignment of patient goals and preferences. Effective communication is needed not only between patients and clinicians but also between health care professionals (table 3). (See 'Shared decision making' above.)

- **Barriers** – Several common barriers to deprescribing such as patient reluctance, care shared between multiple providers, challenges in recognizing appropriate medications, and clinical inertia can be addressed through communication, education, and other strategies. (See 'Common barriers to deprescribing and strategies for overcoming them' above.)

- **Resources** – There are a large variety of resources and tools that have been developed to support health care professionals to deprescribe, such as generic frameworks and drug-specific deprescribing guidelines. (See 'Resources to support deprescribing' above.)

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Topic 122694 Version 22.0

GRAPHICS

Table 1: Patient characteristics and medications that may be high-value targets for deprescribing

Patient characteristics
Use of multiple medications (polypharmacy)
Multimorbidity
Kidney impairment
Multiple prescribers
Transitions of care
Medication nonadherence
Limited life expectancy
Frailty
Dementia
Medications
Opioids for chronic non-cancer pain
Gabapentinoids
Antipsychotics for behavioral changes in patients with dementia
Strongly anticholinergic medications (eg, first-generation antihistamines)
Benzodiazepines
Benzodiazepine receptor agonists
Sulfonylureas
Insulins in patients who could reach appropriate glycemic targets with less or safer glucose-lowering therapy
Chronic use of proton pump inhibitors without strong indication
Chronic use of NSAIDs without strong indication
Aspirin for primary prevention of cardiovascular disease in older patients

NSAIDs: nonsteroidal antiinflammatory drugs.

Graphic 126484 Version 3.0

* [1-6]

Table 2: Medications that can result in adverse drug withdrawal reactions if stopped abruptly

Medication	Symptoms that can result from abrupt discontinuation
Acetylcholinesterase inhibitors	Abrupt cognitive decline, aggression, agitation, hallucinations, reduced consciousness
Antiseizure medications	Anxiety, depression, seizures
Antidepressants	Akathisia, anxiety, chills, gastrointestinal distress, headache, insomnia, irritability, malaise, myalgia
Antiparkinsonian agents	Hypotension, psychosis, pulmonary embolism, rigidity, tremor
Antipsychotics	Dyskinesias, insomnia, nausea, restlessness
Baclofen	Agitation, anxiety, confusion, depression, hallucinations, hypertonia, insomnia, mania, nightmares, paranoia, seizures
Benzodiazepine receptor agonists	Agitation, anxiety, confusion, delirium, insomnia, seizures
Beta blockers	Angina pectoris, anxiety, myocardial infarction, severe hypertension, tachycardia
Clonidine	Agitation, headache, palpitations, severe hypertension
Corticosteroids	Anorexia, hypotension, nausea, weakness
Opioids	Abdominal cramping, anxiety, chills, diaphoresis, diarrhea, insomnia, restlessness
Proton pump inhibitors	Gastric upset, heartburn (due to rebound hyperacidemia)

* This is a partial list. When in doubt, it is typically safer to slowly taper a medication than to stop it abruptly. Adverse drug withdrawal reactions (ADWRs) refer to adverse physiological changes caused by medication withdrawal beyond a simple return of the underlying condition the medications are intended to treat, for example as may occur when medication exposure has resulted in up- or down-regulation of end-organ receptors that result in receptor over- or under-activation when the medication is abruptly withdrawn. Not included in this table is the wider spectrum of adverse drug withdrawal events (ADWEs), which can include return or worsening of the condition a medication was being used to treat or adverse psychological responses to medication withdrawal.[6]

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Graphic 126485 Version 4.0

Table 3: **Points for communication about deprescribing**

Communication with patients (and caregivers/family) to inform deprescribing	▪ Obtain information about experiences with medications (side effects, effectiveness), preferences, and goals of care ▪ Discuss realistic treatment goals – what do they value the most? ▪ Elicit fears about deprescribing
Communication with patients (and caregivers/family) to achieve shared decision-making about deprescribing	Why? ▪ Discuss why deprescribing of specific medications is being recommended – lack of benefits, risk of harm, etc (tailored to the individual and their goals of care) ▪ Frame as "not giving up on you" but as optimizing care ▪ Explain that we may not have good evidence on the benefits and harms of medicines after long-term use
	How? ▪ Frame deprescribing as a trial, with reassurance that the medication(s) can be restarted if necessary ▪ Offer alternative strategies to control bothersome symptoms ▪ Plan tapering/withdrawal process ▪ Discuss monitoring plan
	Fears: ▪ Attend to emotion ▪ Explore and address fears and concerns about deprescribing ▪ Discuss what to do if symptoms occur (monitoring plan)
Communication with other health care professionals	▪ Activate patient to be custodian of medication list (but responsibility should not be solely placed on the patient) ▪ Communicate plan clearly to other clinicians ▪ Collaborate with other clinicians • Don’t be afraid of stopping/making a recommendation about stopping a medication that is determined to be inappropriate • Avoid devolving of responsibility – don't expect others to take responsibility for proactive deprescribing

Graphic 126486 Version 1.0

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